

Ishan Voleti

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Education

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Engineering

Expected Graduation: May 2028

- **Concentration:** Computing Hardware & Emerging Architectures and Distributed System & Software Design
- **GPA:** 3.73 | Eta Kappa Nu (IEEE-HKN) Electrical & Computer Engineering Honor Society
- **Relevant Coursework:** Data Structures & Algorithms, Hardware/Software Systems Programming, Digital Logic & Design, Circuit Analysis, FPGA/VHDL Digital Design Lab, Discrete Mathematics, Electricity & Magnetism

Experience

Trustworthy Robotics Lab – Georgia Tech

Atlanta, GA

Undergraduate Research Scholar

August 2026 - Present

- Building a robust C++ Model Predictive Control solver for real-time quadruped locomotion under disturbance
- Porting solver bottlenecks to CUDA, identifying parallelizable ops and benchmarking power per update
- Investigating FPGA acceleration of solver stages using fixed-point arithmetic and pipelining for a Unitree Go2

Zhou Lab – Drexel University

Philadelphia, PA

Robotics Research Assistant

April 2026 - August 2026

- Automated 360° video-to-Gaussian-Splatting pipeline via COLMAP+3DGRUT, cutting fisheye pose errors by 35%
- Batch-extracted and undistorted 360° frames into perspective views with FFmpeg, standardizing COLMAP input
- Auto-masked pedestrians from training frames using SAM 2 segmentation, cutting reconstruction floaters by 65%

RoboJackets – RoboCup Division

Atlanta, GA

Firmware Engineer

August 2025 - May 2026

- Implemented pulsed signal processing in Rust to improve ball presence detection, reducing false positives by 88%
- Configured RTIC to schedule microsecond-precision tasks, ensuring <15us actuation latency for peak kick power
- Tuned motorboard MOSFET dead-time in firmware, cutting shoot-through spikes by 25% and switching losses

Projects

Autonomous Marine Vehicle | ROS 2, Python, OpenCV, PCB, Raspberry Pi

January 2026 - May 2026

- Engineered a fully autonomous surface vehicle using ROS 2 nodes, achieving <2.5m cross-track error using GPS
- Fused RPLIDAR and computer vision for obstacle avoidance, detecting goal gates and buoys at 95% success rate
- Modified a custom PCB's power regulation for 5V/12V logic in KiCad, interfacing IMU/GPS via I2C/UART

ASIC Design & Verification (SiliconJackets) | SystemVerilog, RTL Design

January 2026 - May 2026

- Designed a 64-bit calculator in SystemVerilog, implementing a custom FSM to sequence multi-cycle additions
- Constructed a modular verification environment (Driver/Monitor/Scoreboard) to validate functional correctness
- Wrote constrained-random tests and SystemVerilog Assertions to expose edge cases, achieving 98% coverage score

Buzz – Embedded Side-Scrolling Game | C, ESP32-C6, ESP-IDF/FreeRTOS

January 2026 - May 2026

- Devised a 97-bucket chained hash table in C to index a sparse, scrolling 52x9 tile map for O(1) collision lookups
- Interfaced a uLCD, joystick, and LED over UART, GPIO, and SPI on an ESP32, driving audio via GPTimer ISR
- Scheduled a real-time 100ms game loop under FreeRTOS using vTaskDelayUntil for deterministic 10Hz execution

Loop (1st Place, Solana Track @ UGAHacks XI) | Rust, Solana, Next.js, Supabase, AI

February 2026

- Architected a trustless scavenger hunt on Solana using Anchor (Rust) and PDAs for automated prize management
- Deployed Haversine-based GPS validation (<100m threshold) and synced real-time player positions via Supabase
- Developed an AI route generator using Google Gemini to create dynamic, localized quests with custom difficulty

Technical Skills

Languages: C, C++, ROS, ROS2, Rust, Python, SystemVerilog, Verilog, VHDL, Java, JavaScript, MATLAB

Digital Design: RTL Design, UVM, SystemVerilog Assertions, Constrained-Random Verification, Cadence Xcelium

Embedded: FreeRTOS, SPI/I2C/UART/GPIO, STM32, Oscilloscope, ARM/ESP32, KiCad PCB, Linux, Git, CMake

Robotics: Nav2, SLAM, Sensor Fusion (IMU/LiDAR/GPS), Isaac Sim, Gazebo, MuJoCo, PyTorch/TensorFlow, CUDA, OpenCV, Docker/CI-CD, Controls/Kinematics/Dynamics, Vision-Language-Action (VLA)

Affiliations: Opportunity Research Scholars, Startup Exchange (Hackathon Team, Sponsorship Lead), Alpha Kappa Psi